

The angle between a face and the base

WORKSHEET



Interactive version

Name _____

Class _____

Date _____

WORKED EXAMPLES

Three worked examples

Example 2 • Given the angle, find the height

TASK

A regular quadrilateral pyramid has base side 10. The angle between a side face and the base is 60° . Find the height.

THE ANGLE

$\varphi = \angle SMO = 60^\circ$ and $OM = 5$.

TRIANGLE SOM • YOUR TURN

Example 3 • Given the lateral edge

TASK

A regular quadrilateral pyramid has base side 6 and lateral edge $SB = 5$. Find the angle between a side face and the base.

HEIGHT • YOUR TURN

TRIANGLE SOM • YOUR TURN

CHECK

Check yourself

The figure shows the same pyramid: base side 4, height $SO = 3$.

Level 2 • satisfactory

1.

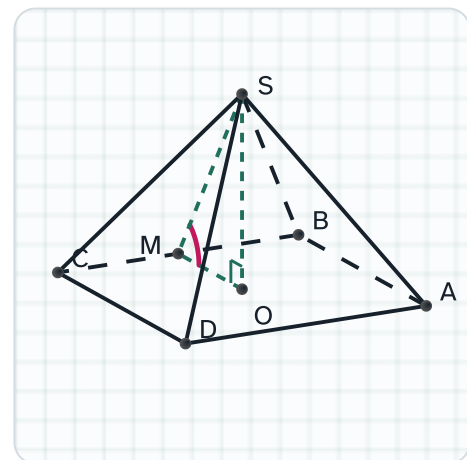
Which angle is the angle between the lateral edge **SB** and the base?

- ☐ A $\angle SMO$
☐ B $\angle SBO$
☐ C $\angle BSO$
☐ D $\angle SBC$

Level 2 • satisfactory

2.

The angle between the face **SCD** and the base is measured at one point of the edge CD. Tap that point on the figure.



3.

Find the length of the apothem SM. Give the exact value or round to two decimal places.

Answer :

4.

Find the angle φ between a side face and the base, in degrees, rounded to 0.1° .

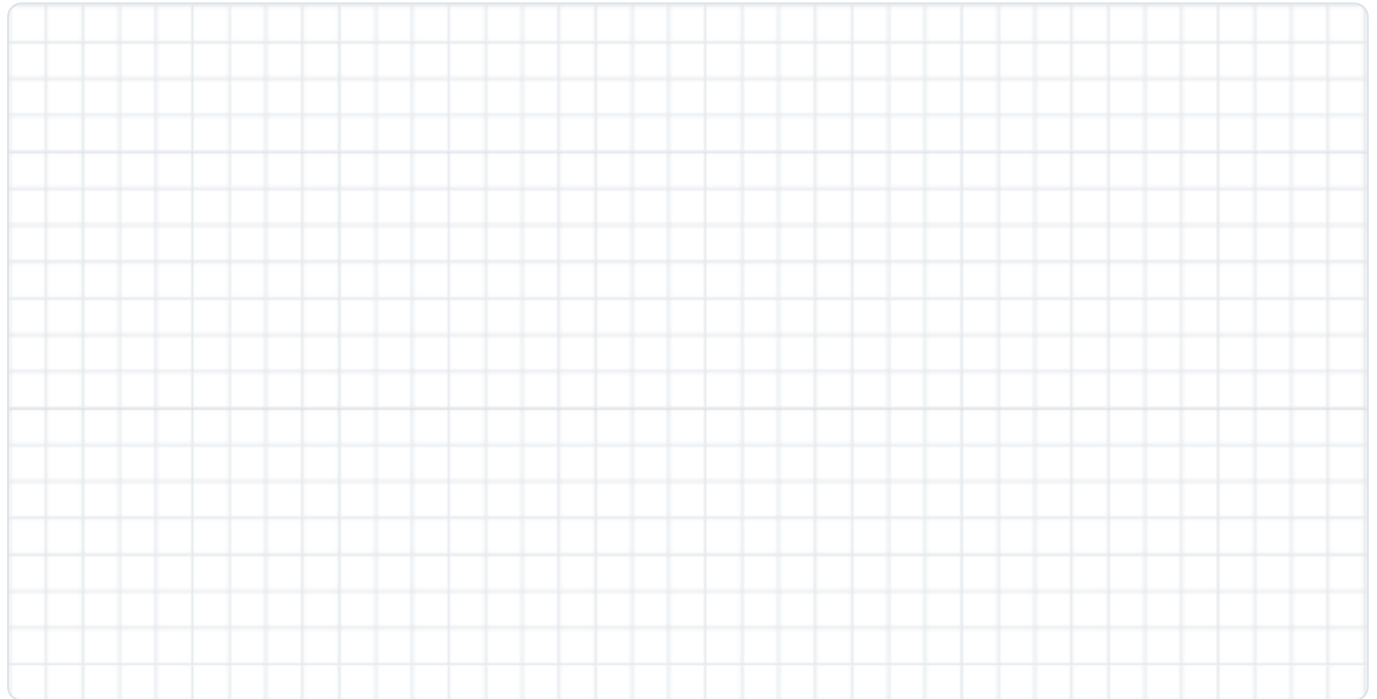
Answer :

EXAM TASK

A regular pyramid, VBE style

The base of a regular quadrilateral pyramid is a square with side 8. The angle between a side face and the base is 60° .

1. Show that the height of the pyramid is $4\sqrt{3}$.
2. Find the lateral surface area of the pyramid.
3. Find the angle between a lateral edge and the base, to 0.1° .



☐ I have done this lessonIt ticks itself when you finish the check.[Unit 3 lessons](#)[Found a mistake?](#)[Next: Distances in space →](#)